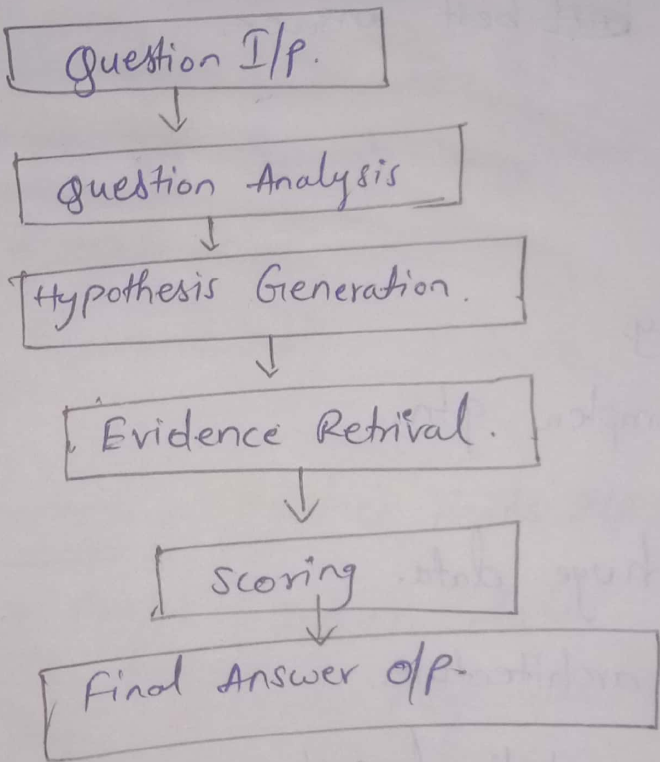
Unit-V

Deep QA Architecture:

⇒ It is the architecture used in IBM watson to answer questions in natural language.

- It is designed to : - understand questions
  - search large data.
  - Give accurate answers.

Diagram :



(i) **Question Analysis :**

- understands the question
- finds keywords & type.

ex: "Who is the president of India?"

(ii) **Hypothesis Generation :**

- Generates possible answers.

ex: "Droupadi Murmu", "Narendra Modi".

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(iii) Evidence Retrieval :

- searches data sources (doc, DB).
- collects supporting info

(iv) Scoring (Evaluation) :

- checks each answer.
- Gives score based on correctness.

(v) Final Answer selection :

- chooses the ~~best~~ best answer.
- Gives o/p.

Advantages :

- High accuracy
- Handles complex qtn.

Limitations :

- Requires huge data.
- Complex architecture.
- High computational cost.

Applications :

- Medical diagnosis sys.
- Chatbots.
- virtual assistants.

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UIMA (unstructured info management Architecture).

- It is developed by IBM.
- used in sys like IBM watson.
- It is a frameworks that helps:
    - Break unstructured data into parts.
    - Analyze it step by step.
    - extract useful info.

Ex: Sentence → words → meaning → useful data.

- It handles unstructured data.

Working: ① I/p data (text, speech)

② process through different components.

③ Extrat info

④ Stores results.

Components:

① collection Reader: Reads I/p data

② Analysis Engine: process data.
- finds keywords, Meanings.

③ CAS (common Analysis Structure):
- Stores processed data.
- shares info b/w components.

④ consumer: uses final results.

ex: display o/p.

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Dg :-

I/p data
↓
Collection Reader
↓
Analysis Engine
↓
CAS
↓
Consumer (o/p).

Advantages :-

- Efficient data processing.
- easy to entend.
- supports large data.

Limitation :-

- complen setup.
- Needs domain knowledge.

Applications:-

- Question answering sys.
- Tent mining.
- chatbots.
- Search engines

Structured knowledge :-

- The data arranged in a clear structure.
- organized using tables, schemas, or graphs.

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Ex:

Name	Age	City
Ram	20	Delhi

⇒ These are orgnized format.

- Easy to search & process
- follows rules (schema).

Types :- (i) Relational DB :- - Data stored in tables
Ex: Sql DB.

(ii) Knowledge Graphs :-
- Data stored as nodes & relationships.
ex: "paris" → capital of → "France"

(iii) ontologies :- - Defines concepts & relationships.
- used in semantic webs.

Working :-
1. Data is collected.
2. organized into structure.
3. Stores in DB/graph
4. Used for queries & reasoning.

Role in CC :- - Helps machines understand info clearly.
- Improves accuracy of AI sys.

Advantages :-
- Easy to access
- Fast processing
- Accurate results

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Limitations :- Difficult to create Structure.
- Less flexible

Applications:
- Search engines
- Business DB
- AI assistants.
- Q/A systems.

Business Implications :-

- It means how CC affects businesses & their operations & decisions.

- The CC helps businesses:-
  - understand data
  - Predict future trends
  - Automate tasks.

Business ~~Impacts~~ Impacts :-

① Better Decision Making :-
- AI analyzes large data.
- Give accurate insights.

ex:- predicting sales.

② Automation :- - Replaces manual works.
- Saves time & effort.

ex:- chatbots for customer support.

③ personalization :- provides customized services.

ex: product recommendations.

④ Cost Reduction :- Less human effort & efficien operatio

⑤ Risk mangement :- - Detects fraud & errors
- predicts risks.

Business Areas affected :-

- Marketing, finance, Healthcare, Retail, Banking etc...

Advantages :- - faster decisions.
- Better accuracy

Challenges :-
- High cost
- Data privacy issues
- Dependency on technology.

Applications :-
- Recommendation sys
- fraud detection
- virtual assistants.

Building cognitive Applications :-

- It means creating s/w sys that can:-
  - understand data
  - Learn from experience
  - Makes decisions like humans.

- These applications use AI techniques such as:-
  - ML
  - NL
  - Data Analytics.

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Steps to build cognitive Applications

① Define the problem :- Identify what the sys should do.

ex:- Customer Support Chatbot.

② Collect Data :- Gather structured & unstructured data.

ex:- Customer queries.

③ Data processing :-
- Clean & organize data.
- Convert it into usable form.

④ Choose Model / Technique :-
- Select suitable AI method.

ex:- NLP for text. , ML for predictions.

⑤ Training, Testing & Evaluation of model :-
- Train sys using data.
- Check performance & improve accuracy.

⑥ Deployment :-
- Launch application for users.

⑦ Continuous Learning :-
- update sys with new data.
- Improve over time.

Tools & technologies :-
- ML frameworks
- Cloud platforms.
- API's

Ex: IBM Watson tool

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Advantages:- - Automates task
- Improves accuracy
- Enhances user experience.

Challenges:-
- Needs large data.
- Complex design.
- High cost.
- Data privacy issues

Applications:-
- Chatbots
- virtual assistants
- Health care sys
- fraud detection

Applications of Cognitive Computing & Sys:-

- Cognitive computing sys are AI based sys that can:- . learn from data.
  . understand language
  . Makes decisions.

- They are used in many real world areas to solve complex problems.

Applications are:-

① Healthcare:- - Diagnose diseases
- Suggest treatments.

ex:- IBM watson helping Doctors.

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(2) Eduction :- smart learning sys
- personalized teaching.

(3) Banking & Finance :-
- fraud detection
- Risk analysis
- Customer support.

(4) customer service :-
- Chatbots & virtual assistants.
- 24/7 support.

(5) Transportation :-
- Traffic prediction.
- Autonomous vehicles.

(6) security & cyber security :-
- Threat detection
- Data protection.

Types of cognitive systems used :-
- Q/A sys
- Recommendation sys
- Speech recognition sys
- Image " "
- Decision support "

Benifits :- Faster Decision making
- High accuracy.

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- Automation of tasks

Challenges :-
- High cost
- complex
- Data privacy issues
- Required skilled people

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